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Project Report

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iOS Artifact Analyzer

A Python-Based Mobile Forensic Triage Tool

Digital Forensics — Semester Project

1. Abstract

This project presents the design and implementation of the iOS Artifact Analyzer a Python-based digital forensics tool capable of extracting, parsing, and presenting forensic artifacts from Apple iOS device backups.

The tool targets iTunes/Finder-generated backup files and employs multiple extraction techniques including direct SQLite database querying for live artifacts and binary-level SQLite freelist page carving for deleted data recovery.

The analyzer extracts seven categories of forensic artifacts:

- SMS and iMessage records
- Call logs
- Contacts
- Safari browsing history
- Photo metadata with GPS coordinates
- Installed application lists
- Device location history

Additionally, the tool implements a two-stage deleted message recovery pipeline using SQLite WAL scanning combined with freelist page binary carving.

All findings are presented in a professionally styled HTML report with interactive GPS mapping, chain-of-custody documentation, and SHA-256 hash verification.

2. Introduction

2.1 Background

Mobile devices have become the primary digital artifact in modern criminal investigations.

iPhones are especially important because they store a large amount of forensic evidence including:

- Messages
- Photos
- GPS locations
- Browser history
- Contacts

- Application data

iTunes backups provide investigators with a legally accessible source of forensic evidence without jailbreaking the device.

2.2 Problem Statement

Commercial forensic tools like:

- Cellebrite UFED
- Oxygen Forensic Detective
- MSAB XRY

are extremely expensive and inaccessible for many students and educational institutions.

Therefore, this project aims to build an open-source educational forensic tool for iOS backup analysis.

2.3 Objectives

The objectives of the project are:

- Extract forensic artifacts from iOS backups
 - Recover deleted messages using SQLite carving
 - Generate professional forensic reports
 - Verify evidence integrity using SHA-256 hashing
 - Analyze GPS location data
 - Study iOS forensic limitations
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3. Literature Review

3.1 iOS Backup Structure

iOS backups contain device data in a structured format.

Important files include:

- Manifest.db
- Info.plist
- SQLite databases

The backup system maps original file paths to hashed filenames.

3.2 SQLite Forensics

SQLite databases store most mobile forensic artifacts.

Deleted records may still remain inside:

- Freelist pages
- WAL files

These artifacts can be recovered using forensic carving techniques.

3.3 Commercial vs Open Source Tools

Tool	Type	Open Source
Cellebrite UFED	Commercial	No
Oxygen Forensic Detective	Commercial	No
MSAB XRY	Commercial	No
libimobiledevice	Open Source	Yes
iLEAPP	Open Source	Yes
iOS Artifact Analyzer	Open Source	Yes

4. Methodology

4.1 System Architecture

The system follows four main stages:

1. Acquisition
2. Parsing
3. Extraction
4. Reporting

4.2 Backup Parsing

The tool reads:

- Manifest.db
- Info.plist
- SQLite artifact databases

It maps hashed filenames back to their original locations.

4.3 Artifact Extraction

SMS Extraction

Messages are extracted from sms.db using SQLite queries.

The tool supports:

- SMS
- iMessage
- Contact mapping
- Timestamp conversion

Call Logs

Call history is extracted from:

- CallHistory.storedata

Photo GPS Extraction

GPS coordinates are extracted from:

- Photos.sqlite

This helps investigators identify where photos were taken.

4.4 Deleted Data Recovery

SQLite Freelist Carving

When SQLite records are deleted, their pages are marked as free but the data may still exist.

The tool scans freelist pages and recovers readable text fragments.

WAL File Scanning

SQLite WAL files may contain deleted records before checkpointing.

The analyzer scans WAL frames to recover deleted messages.

5. Implementation

5.1 Technologies Used

Technology	Purpose
Python	Main logic
sqlite3	Database analysis
plistlib	iOS plist parsing

Technology	Purpose
exifread	Photo metadata
Leaflet.js	GPS maps
SHA-256	Integrity verification

5.2 Project Structure

The project contains multiple modules:

- analyzer.py
- report_generator.py
- sms_extractor.py
- call_extractor.py
- safari_extractor.py
- photos_extractor.py
- location_extractor.py

5.3 Chain of Custody

The system verifies integrity using SHA-256 hashing before and after analysis.

This ensures forensic evidence is not modified.

6. Results and Testing

Extracted Artifacts

Artifact	Status
SMS	Complete
Calls	Complete
Contacts	Complete
Safari History	Complete
Photos + GPS	Complete
Installed Apps	Complete
Deleted SMS	Partial

Deleted Message Recovery Results

The tool successfully recovered deleted SMS records from:

- Freelist pages
- WAL files

Recovery success depends on database activity.

7. Forensic Significance

Investigative Importance

The extracted artifacts help investigators:

- Identify communication patterns
- Track locations
- Recover deleted evidence
- Establish timelines

Legal and Ethical Considerations

The tool should only be used on authorized devices or backups.

Unauthorized forensic analysis may violate cybercrime laws such as:

- Prevention of Electronic Crimes Act

8. Challenges and Limitations

Challenges

- Apple timestamp conversion
- Encrypted backup handling
- iOS schema changes

Limitations

- No physical acquisition
- Requires backup password for encrypted backups
- Limited iOS 17/18 support
- No iCloud backup support

9. Conclusion

The iOS Artifact Analyzer demonstrates that professional-level mobile forensic analysis can be performed using open-source tools.

The project successfully implements:

- iOS backup parsing
- Artifact extraction
- Deleted message recovery
- GPS analysis
- Forensic report generation

The project also highlights the growing challenge of performing forensic analysis on newer iOS versions due to advanced encryption and security protections.